

M15 - Enhancement of the Antibacterial Activity of Bee Pollen Ethanolic Extract by α -Cyclodextrin Inclusion Complexes

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Introduction

Natural Product, a valuable source of biologically active compounds

Polyphenols and flavonoids

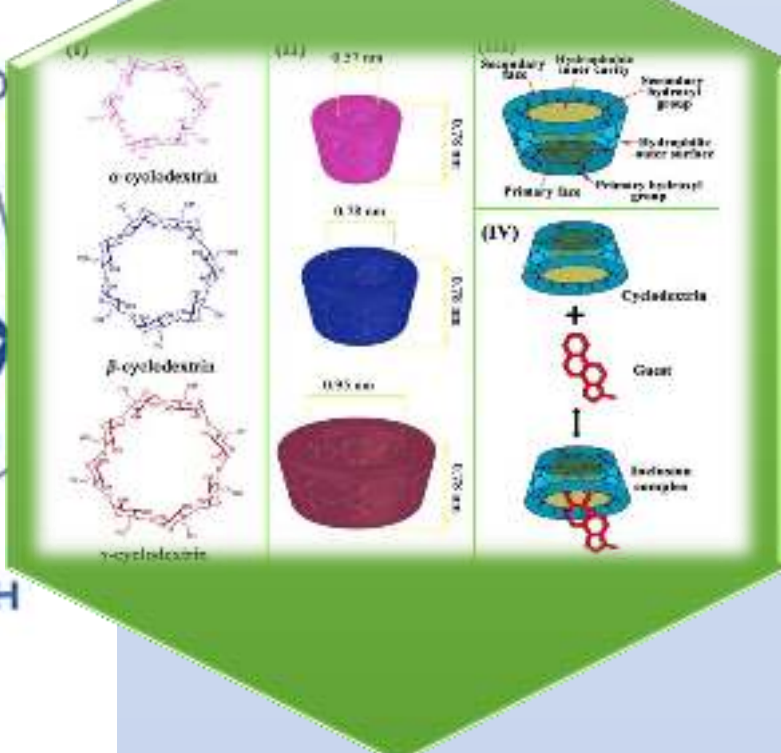
Which are known for their antioxydant and antibacterial activities

Quercetin is able to protect biological system against oxidative stress

Bee pollen

Exhibit poor aqueous solubility and limited stability wich may restrict their bioavailability

Cyclodextrins are cyclic oligosaccharides capable of forming inclusion complexes with hydrophobic molecules. In particular, α -cyclodextrin (α -CD) can encapsulate bioactive compounds within its hydrophobic cavity, potentially improving their solubility, stability and protection against degradation



Materials and Methods

1. Preparation of Bee Pollen Extract

Bee pollen was used as the raw material for the preparation of an extract rich in bioactive compounds. Ethanolic extraction was performed to promote the recovery of phenolic and flavonoid compounds. After extraction, the solvent was removed to obtain the bee pollen ethanolic extract (BPE), which was subsequently used for the different analyses and formulation studies.

2. Determination of Bioactive Compounds

The total flavonoid content (TFC) was determined using a colorimetric method and expressed as quercetin equivalents (QE). The total phenolic content (TPC) was also determined using a colorimetric method and expressed as gallic acid equivalents (GAE). Quercetin was used as a marker compound for the bee pollen extract.

Preparation of Inclusion Complexes BPE/ α -CD inclusion complexes

were prepared using two different methods: Physical mixing (PM): the bee pollen extract and α -cyclodextrin were directly mixed at the appropriate ratio. Kneading method (KN): α -cyclodextrin and the extract were mixed in the presence of a small amount of solvent and kneaded to promote interactions between the bioactive compounds and the α -CD cavity. The resulting formulations were subsequently dried and subjected to physicochemical characterization.

4. Physicochemical Characterization The formation of the inclusion complexes

was investigated using several characterization techniques, including : Fourier-transform infrared spectroscopy (FTIR) to identify potential interactions between the bioactive compounds and α -CD; X-ray diffraction (XRD/XRPD) to evaluate changes in the crystalline or amorphous structure; Thermal analysis, including differential scanning calorimetry (DSC) and thermogravimetric analysis (TGA), to investigate changes in the thermal properties of the formulations following complexation.

5. Antioxidant Activity The antioxidant activity of the extract and formulated systems

was evaluated using the DPPH radical scavenging assay. The ability of the samples to scavenge free radicals was determined and compared among the bee pollen extract, pure quercetin, and the formulated systems.

6. Antibacterial Activity The antibacterial activity

was evaluated using the disk diffusion method against four bacterial strains: Escherichia coli; Klebsiella pneumoniae; Staphylococcus aureus; Bacillus cereus. The diameters of the inhibition zones were measured to compare the antibacterial activity of raw bee pollen, the ethanolic extract, and the α -CD inclusion complexes.

7. Evaluation of the Protective Effect on Spermatozoa The protective effect of the different treatments

was investigated using spermatozoa stored at 4 °C. Sperm motility parameters were monitored using a Computer-Assisted Sperm Analysis (CASA) system. The main parameters evaluated were :VCL: curvilinear velocity; VSL: straight-line velocity; VAP: average path velocity. The results obtained for the different treatments were compared with those of an untreated control group to evaluate their ability to maintain sperm quality and motility during cold storage.

Results and discussions

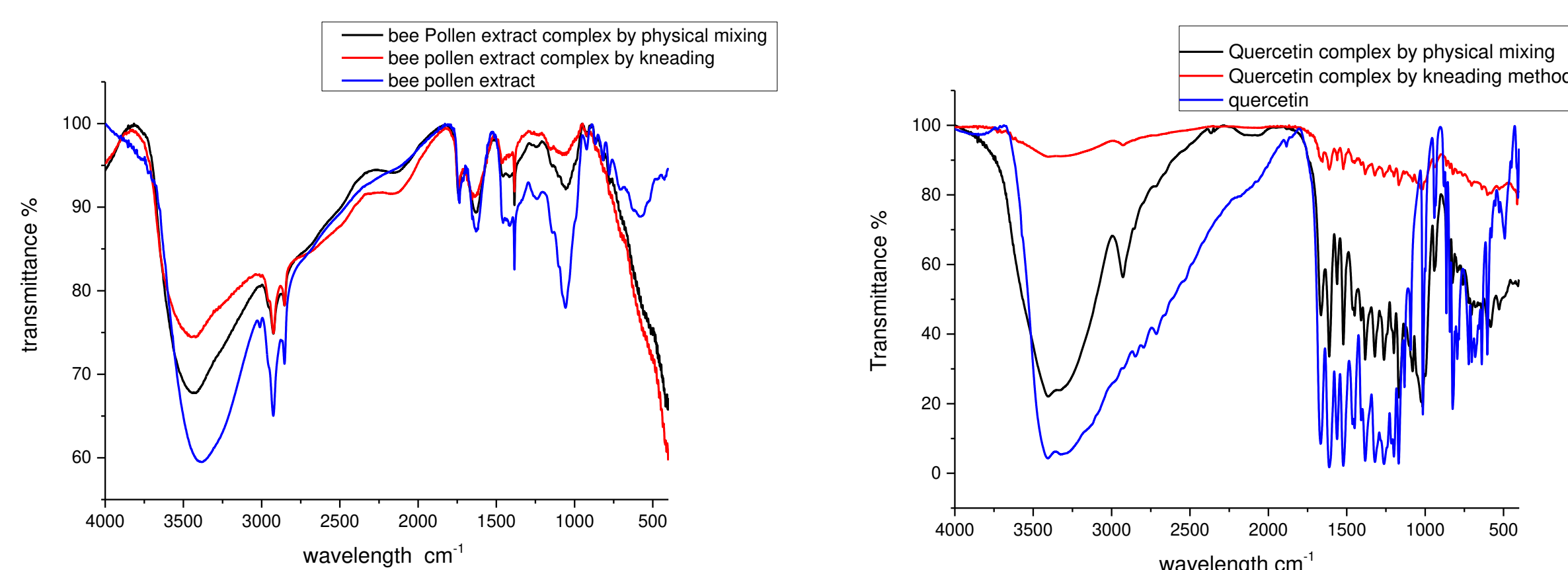


Figure 1. FTIR of α -CD-extract complex. And FTIR of the Quercetin and its complexes

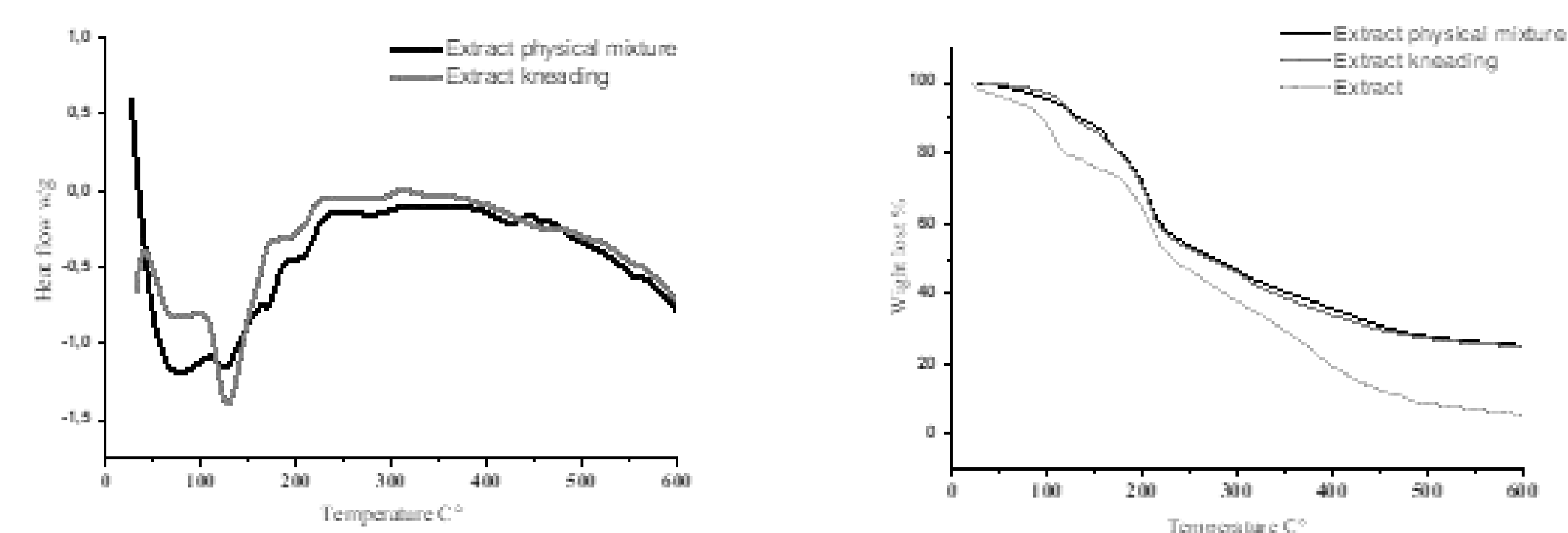


Figure 2. Thermograms (ATG)/(DSC) of the bee pollen extract, bee pollen extract physical mixture and bee pollen extract complex.

Table 1. Inhibition zones (mm) observed for ethanolic bee pollen extract and crude pollen against bacterial strains.

Concentrations of ethanolic extracts of bee pollen	E.C	K.P	S. A	B.C
0,005g/10ml DMSO Extract	8	10	-	7
0,05mg/5ml DMSO Extract	8	11	-	-
0,25g/2,5 DMSO Extract	9	11	10	9
0,01g/ml water extract	11	13	-	-
0,025 g/ml ethanol. extract	11	13	15	10
	12	15	10	9

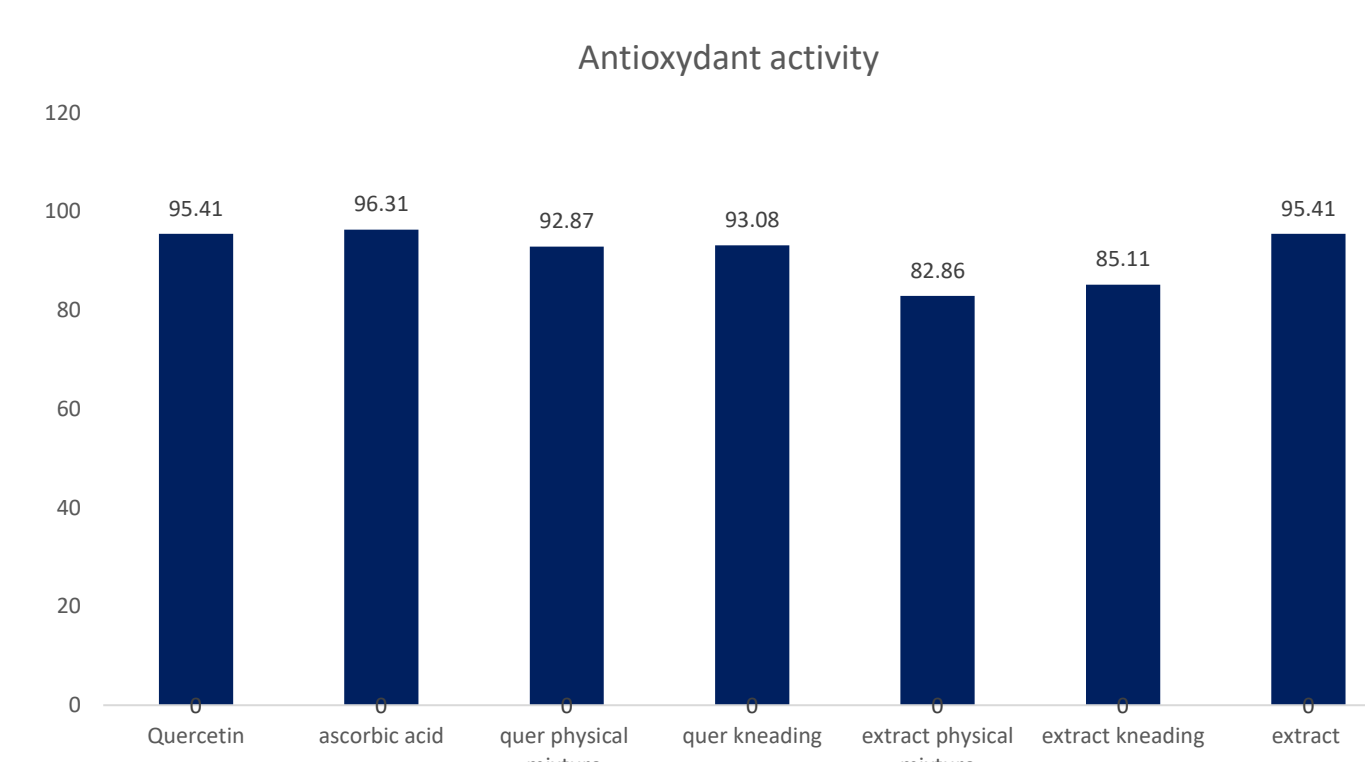


Figure 3. Antioxidant activity of ascorbic acid, quercetin, bee pollen extract, and their inclusion complexes.

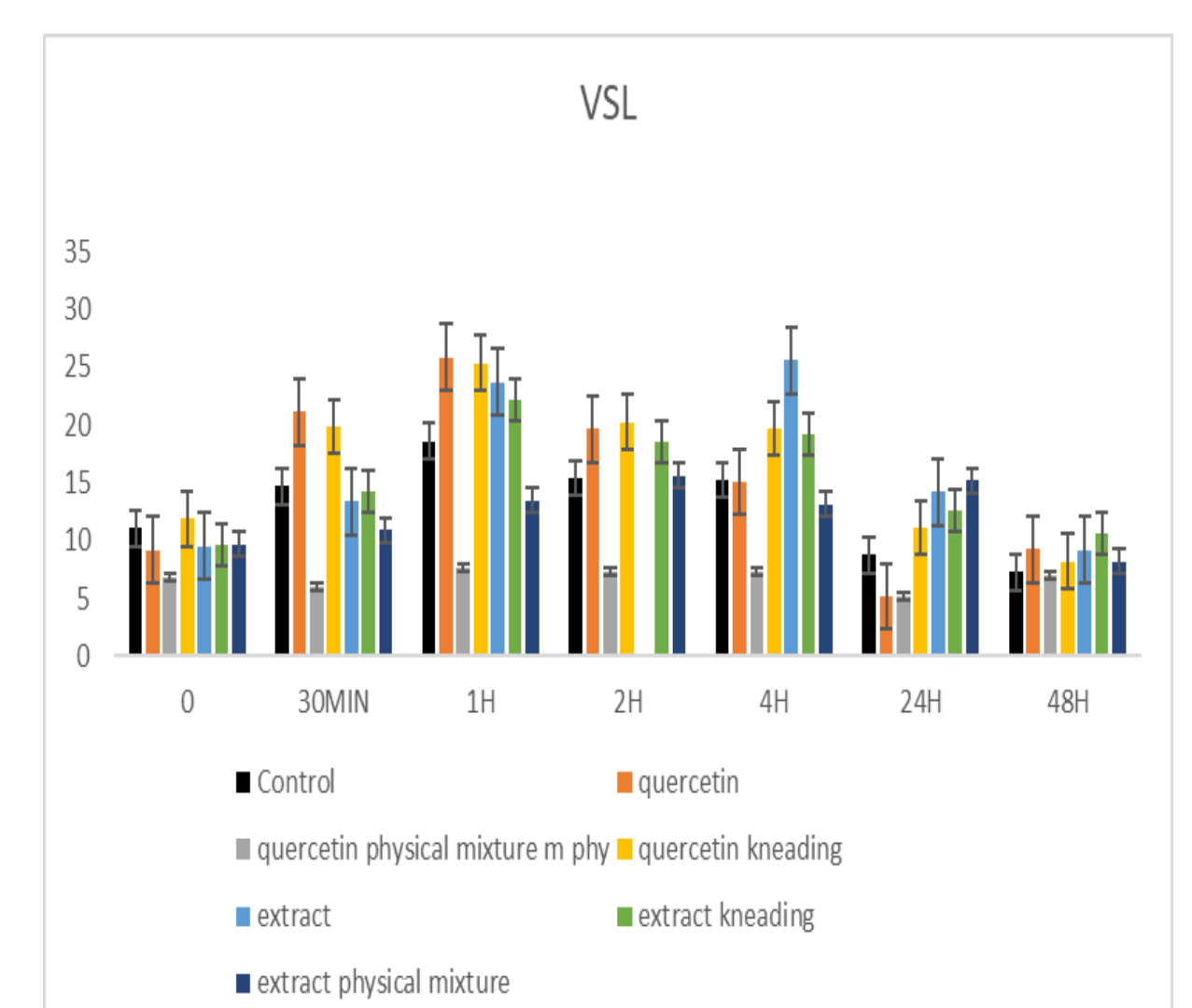


Figure 4. Sperm motility parameter VSL after cryopreservation with bee pollen ethanolic extract, quercetin, and their complexes.

Conclusion

This study demonstrated that bee pollen ethanolic extract is a valuable source of bioactive compounds with significant antioxidant and antibacterial activities. However, their limited aqueous solubility and stability may restrict their practical applications. The successful encapsulation of the bee pollen bioactive compounds within α -cyclodextrin was confirmed by FTIR, TGA, and DSC analyses, which revealed changes in the molecular interactions and thermal behavior of the formulations. Among the investigated systems, the kneading method showed the most promising performance, suggesting more effective interactions between the bee pollen bioactive compounds and the α -CD cavity. Furthermore, α -CD encapsulation enhanced the antibacterial activity of the extract and provided a protective effect on spermatozoa during storage at 4 °C, as evidenced by the better maintenance of VCL, VSL, and VAP parameters. Overall, α -cyclodextrin encapsulation represents a promising strategy for improving the stability and biological performance of bee pollen bioactive compounds, highlighting their potential application as natural additives for sperm preservation and in pharmaceutical and nutraceutical formulations.